

Real-Time IoT-Enabled Water Quality Monitoring and Groundwater Availability Analysis Using Machine Learning

Project ID : W11

Capstone Project Phase 2

ESA Evaluation

Team Composition

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Guide : Prof. NAGARAJ L J

Outline of Presentation

- Introduction & Motivation
- Literature Survey
- Objectives
- Problem Statement
- Methodology
- Work Progress
- Required Hardware / Software details
- Results and Analysis
- Project Deliverables
- References
- Project timeline Gantt Chart

Introduction & Motivation

- Water pollution poses serious threats to ecosystems and public health
- Existing monitoring methods lack real-time data monitoring.
- Integrated approach using sensors to monitor water pollution, analysing the ground level water addressing these shortcomings effectively.
- The urgent need to enhance water quality monitoring drives this research.
- With pollution incidents on the rise, there's a huge demand for innovative solutions that offer real time data.
- Developing sensor networks, this study aims to bridge current monitoring gaps, empowering stakeholders with actionable data to reduce pollution risks.

[1] K. Banerjee et al., "Assessing Water Quality Index Near Industrial Regions and Aiding in Effective Water Management and Controlling Water Pollution Level," 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), Uttar Pradesh, India, 2022, pp. 1987-1991

Author's work	The author emphasizes the correlation between industrial waste discharge and changes in crucial water quality parameters, such pH level, and temperature, highlighting their collective influence on altering the Water Quality Index (WQI)
Inference	• This research presents an IoT-based solution for real-time water pollution monitoring. • The system aims to improve water quality management by providing timely alerts to authorities.
Limitations	• The study does not incorporate machine learning algorithms for advanced data analysis and predictive modeling. • The absence of machine learning techniques may limit the system's ability to provide deeper insights and predictive capabilities for water quality management.

[2] Md. Jahirul Islam, Asaduzzaman, “Smart Water Quality Monitoring and Controlling System”, 5th International Conference on Electrical Information and Communication Technology (EICT), 17-19 December 2021, Khulna, Bangladesh

Author's work	The paper introduces the problem of water contamination monitoring and the need for an IoT-based smart system to measure and alert the water quality parameters in real time.
Inference	The paper introduces an IoT-based water quality monitoring and control system, leveraging various sensors to monitor parameters like flow, pH, turbidity, and temperature in real-time. It highlights the system's ability to automate processes such as tank cleaning and send alerts for substandard water quality. Experimental results demonstrate a promising accuracy of 93% in detecting water quality issues. The system is cost-effective, utilizing low-cost components, with potential for further enhancement using industrial-grade sensors. Overall, the paper underscores the significance of IoT technologies in addressing global water quality challenges.
Limitations	One limitation is the potential accuracy compromise due to the use of low-cost components, while environmental factors like weather may affect sensor performance. Additionally, scalability and adaptation to diverse water quality standards and cybersecurity vulnerabilities are unaddressed.

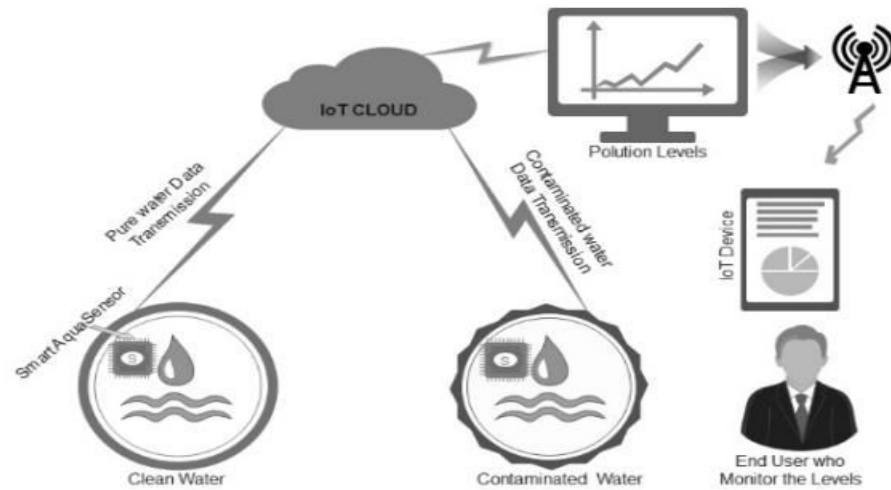


Fig. 1. Smart Water pollution Monitoring Systems with Smart Aqua sensors and IoT devices.

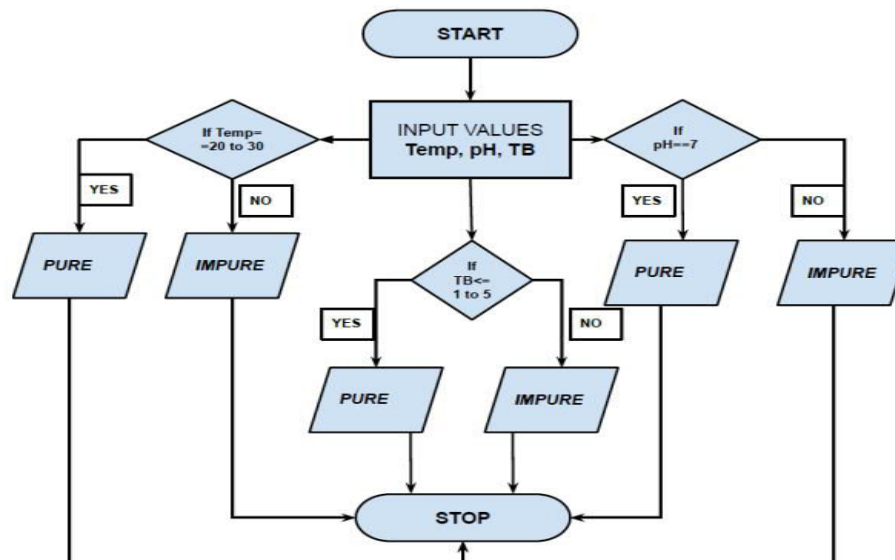


Fig. 2. System flow diagram

[3] M. N. Vamsi Thalatham, P. Lanka and J. N. V. R. S. Kumar, "An IoT Based Smart Water Contamination Monitoring System," 2023 International Conference on Intelligent Systems for Communication, IoT and Security (ICISCoIS), Coimbatore, India, 2023, pp. 387-391

Author's work	The author proposes a system that leverages Internet of Things (IoT) technology to monitor water quality and quantity in real-time.
Inference	The paper presents a real-time water contamination monitoring system leveraging IoT and Embedded Systems to address challenges in urban water supply maintenance due to population growth and pollution. By measuring turbidity, pH, and temperature, the system aims to estimate water quality, offering efficiency and accuracy. Utilizing low-cost components and cloud-based analysis, it provides a cost-effective solution with potential scalability for widespread implementation. However, potential limitations include accuracy compromises with low-cost components and unaddressed cybersecurity vulnerabilities. Future enhancements could involve incorporating additional sensors for comprehensive water quality assessment, making it a promising solution for smart city applications.
Limitations	The main limitations of the proposed system include potential compromises in accuracy due to the use of low-cost components, particularly in comparison to industrial-grade sensors. Additionally, the system may be vulnerable to cybersecurity threats, especially when data is transmitted over the Internet. These limitations suggest the need for careful consideration of component quality and cybersecurity measures in future iterations to ensure reliable and secure operation.

[4] K. S., S. T.V., M. S. Kumaraswamy and V. Nair, "IoT based Water Parameter Monitoring System," 2020 5th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2020, pp. 1299-1303

Author's work	The authors highlighted a three-layered approach for water monitoring, utilizing sensors for data acquisition, Gsm and Ethernet for transmission, and IoT platforms for processing.
Inference	-EXO Sonde will collect the data and monitoring will be done at server-side -Data from the primary controller is stored, then transmitted to the server via GSM, without real-time data due to occasional GSM connection issues.
Limitations	-The Exo Sonde is very expensive and lack of some other parameters which are affecting the quality of water -Once the data will be transmitted to the server. The timestamp will be removed. The transmission of data to the server is done through GSM. The GSM connection can't be achieved every time, hence the real time values are not always available.



Fig .2. EXO Module

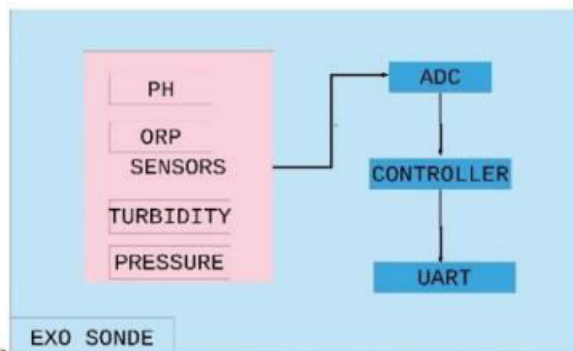


Fig .3. Architecture of EXO Sonde

The smart sensor options in the EXO are conductivity, dissolved oxygen content, pH, rhodamine, total algae, and turbidity

III. SYSTEM OVERVIEW

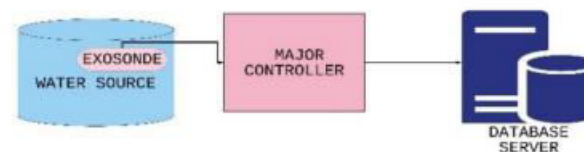


Fig .1. System Overview

Result on server side

WATER PARAMETER MONITORING				
STN_15_2020017000532_EXO				
Node voltage = 1.5 volts				
PRESSURE (Pascal)	TEMPERATURE (Degree celsius)	PH (no unit)	ORP (milli volt)	TURBIDITY (FTU)
-7841	28.807	9.47	410.05	0.31
-7464	28.812	9.47	410.02	0.30
-7932	28.806	9.47	410.00	0.30
-7971	28.818	9.47	409.97	0.31
-7427	28.815	9.47	409.95	0.27
-7806	28.813	9.47	409.99	0.30
-7865	28.818	9.47	409.86	0.30
-9124	28.817	9.47	409.80	0.28
-8708	28.823	9.47	409.75	0.29
-7683	28.536	9.31	422.29	0.29

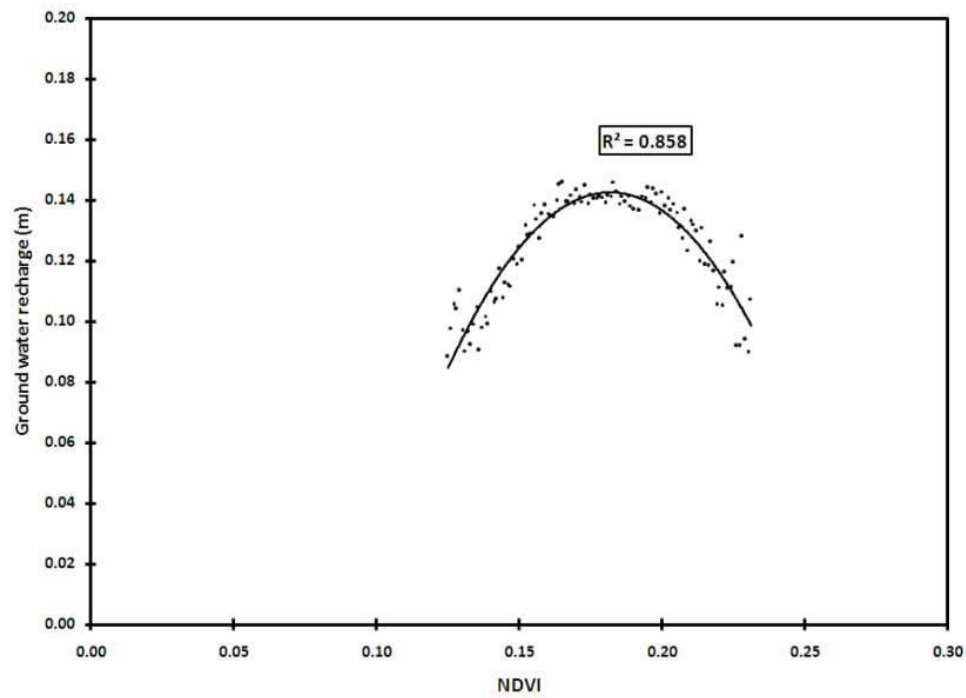
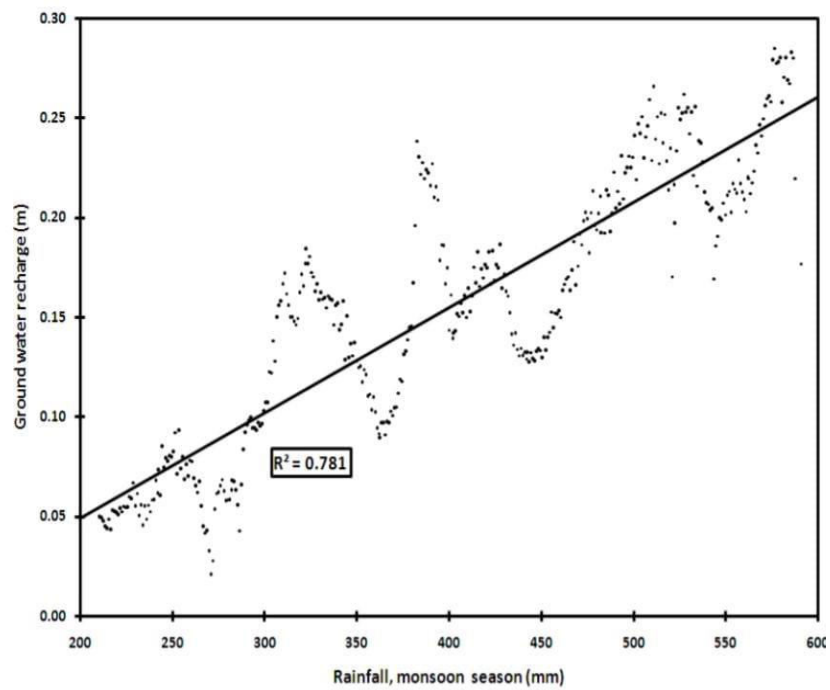
Fig .8. Monitoring server

[5] N. Iqbal et al., "Groundwater Level Prediction Model Using Correlation and Difference Mechanisms Based on Boreholes Data for Sustainable Hydraulic Resource Management," in IEEE Access, vol. 9, pp. 96092-96113, 2021, Doi: 10.1109/ACCESS.2021.3094735.

Author's work	The authors of the paper have collectively developed an ensemble machine learning model to predict groundwater levels, aiming to enhance the efficiency of borehole drilling and sustainable management of hydraulic resources.
Inference	The paper details a machine learning model for predicting groundwater levels, which could revolutionize groundwater management. It also shows that proper data handling improves prediction accuracy, influencing future groundwater studies.
Limitations	Data Dependency: The model's accuracy is highly dependent on the quality and comprehensiveness of the input data. Generalizability: The model may not perform as well when applied to different geographical areas or datasets not represented in the training data.

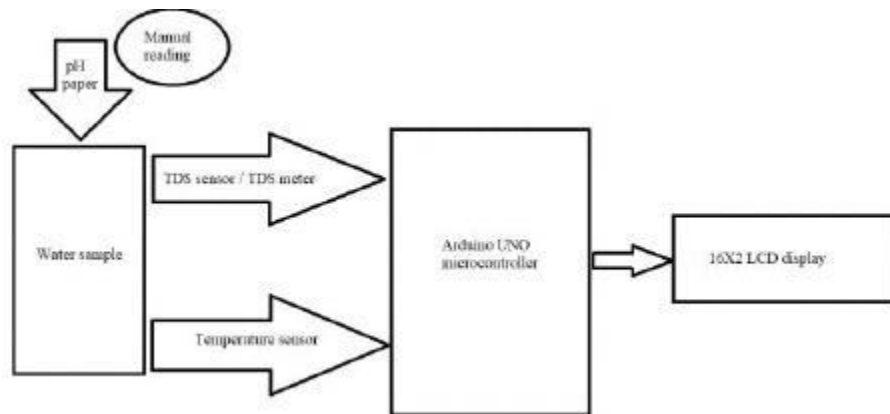
[6] European Journal of Sustainable Development A methodology based on spatial distribution of parameters for understanding affect of rainfall and vegetation density on groundwater recharge Vijai Singhal and Rohit Goyal

Author's work	The authors aim to understand the relationship between rainfall, vegetation density, and groundwater recharge.They utilize spatial distribution analysis to explore how these factors impact groundwater replenishment.
Inference	The study reveals that groundwater recharge increases with Normalized Difference Vegetation Index (NDVI) up to a certain point.Beyond that threshold, further vegetation density does not significantly enhance recharge.Additionally, the research identifies a linear correlation between groundwater recharge and rainfall
Limitations	Simplification. Data Availability Generalization Temporal Variability



[7] S. H. Priyadarshini, P. S., R. P. B., V. K. V. and A. D. V A, "AQUASENSE: Sensor Based Water Quality Monitoring Device," 2023 International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), Erode, India, 2023, pp. 1786-1789

Author's work	Parameters Measured: The device collects water samples from different regions in Bangalore city and measures ph,turbidity
Inference	Temperature Influence: The temperature affects TDS measurements due to its impact on solubility. Affordability and Efficiency: AQUASENSE is an affordable and efficient alternative to existing water quality monitor
Limitations	Local Context: The device's effectiveness may vary based on the specific water sources and environmental conditions in Bangalore. Calibration: Regular calibration is essential to maintain accuracy.



[8] "Study on reciprocal relationship among water amount-water quality-water efficiency based on the SWAT_WAQR model "2021 7th International Conference on Hydraulic and Civil Engineering & Smart Water Conservancy and Intelligent Disaster Reduction Forum (ICHCE & SWIDR)

Author's work	work involves developing and applying the SWAT_WAQR model to analyze the interplay between water consumption, efficiency, and river water quality in the Yulin Basin.
Inference	How changes in water consumption and pollution emissions impact river water quality. By using the SWAT_WAQR model. The authors simulate different water conservation scenarios in the Yulin catchment. The model predicts the effects of economic water consumption and pollutants on water quality.
Limitations	Model Assumptions Simplifications Local Context External Factors



[9] M. Jafril Alam, S. Kar, S. Zaman, S. Ahamed and K. Samiya, "Forecasting Underground Water Levels: LSTM Based Model Outperforms GRU and Decision Tree Based Models," 2022 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE), Naya Raipur, India, 2022, pp. 280-283. doi: 10.1109/WIECON-ECE57977.2022.10151230.

Author's work	The authors of the study on forecasting underground water levels using machine learning and deep learning. However, the research highlights the significance of these models in addressing water depletion and emphasizes the superior performance of LSTM-based models for time series forecasting.
Inference	The paper presents a study on forecasting underground water levels using machine learning and deep learning models, with the LSTM-based model outperforming others in accuracy.
Limitations	The paper discusses the effectiveness of various machine learning models for forecasting underground water levels, highlighting that deep learning models, particularly LSTM, outperform traditional algorithms. However, it acknowledges limitations such as the need for more diverse data and the potential development of an attention mechanism-based system for improved forecasting.

[10] S. K. T. K, M. Hanumanthappa, S. K. P. S and H. B. V, "Data Driven Approach to Predict Ground Water Level using Support Vector Regression," 2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC), Salem, India, 2022, pp. 01-06

Author's work	VM-based model predicts groundwater levels and water consumption, aiding in sustainable resource management.
Inference	Geographical and climatic factors utilized to assess available water for diverse domestic needs.Data-driven approach offers insights for effective water allocation amidst growing scarcity and population demands.
Limitations	Reliance on historical data may limit predictive accuracy.Applicability may be restricted by geographical and climatic factors.

Summary of Literature

- Industries depend on water for production and waste disposal, often polluting it with hazardous chemicals. Currently, 80% of wastewater is untreated globally, harming ecosystems and public health. This paper proposes an IoT-based system to monitor water quality and alert authorities to pollution in real-time. It also aids in sustainable water management by categorizing water based on its suitability for various uses. [1]
- This paper proposes an IoT-based system for monitoring and controlling water quality, which analyzes and purifies water using various sensors that transmit data to remote servers. The system automatically manages water pumps and alerts the owner when water purity falls below a threshold, triggering tank cleaning and disposal of dirty water. Data from 750 samples were analyzed to assess correlations among water parameters, achieving a 93% accuracy in detecting water quality. This system effectively monitors and maintains water purity in real-time. [2]

- This paper introduces an IoT-based system for real-time monitoring of residential water quality, using sensors to measure pH, turbidity, and temperature. Data is sent to a cloud server for analysis, providing faster and more cost-effective results than traditional lab testing. The system enables timely alerts to maintenance agencies, improving response to contamination issues. It offers efficient, automated monitoring with high accuracy and reduced resource demands. [3]
- This paper proposes an automated water monitoring system using a multi-parameter sensor node, EXO Sonde, to gather accurate real-time data on water quality. The system addresses challenges in water applications like smart irrigation by processing data through a layered architecture and transmitting it to a server via GSM. Data is stored locally during network loss and visualized on a web interface. This solution enhances reliability in water monitoring and management applications. [4]

- The 2021 study by N. Iqbal and colleagues presents a groundwater level prediction model that uses correlation and difference mechanisms based on borehole data to support sustainable hydraulic resource management. This model aims to improve the prediction accuracy of groundwater levels, enhancing resource management strategies. The study leverages borehole data to identify key patterns and predict groundwater variations.[5]
- The study by Vijai Singhal and Rohit Goyal investigates the influence of rainfall and vegetation density on groundwater recharge using spatial distribution analysis. It finds that groundwater recharge initially increases with vegetation density (measured by NDVI) but plateaus beyond a certain threshold. A linear relationship is observed between rainfall and recharge. However, limitations include simplifications, data availability constraints, generalization issues, and temporal variability.. [6]

- The 2023 study by S. H. Priyadarshini et al. introduces AQUASENSE, a water quality monitoring device that measures pH and turbidity in water samples from Bangalore. Findings indicate temperature influences TDS readings due to solubility effects. AQUASENSE is highlighted as an affordable, efficient alternative to existing water monitors. However, its effectiveness may vary with local water sources, and regular calibration is needed for accuracy. [7]
- This 2021 study applies the SWAT_WAQR model to examine how water consumption, efficiency, and river water quality interact in the Yulin Basin. The model simulates various water conservation scenarios, predicting the impact of economic water use and pollutant emissions on river quality. Findings highlight the effects of water consumption and pollution on water quality, though results are limited by model assumptions, simplifications, local context, and external factors. [8]

- This study compares machine learning and deep learning models for forecasting underground water levels, finding that LSTM outperforms GRU, Random Forest, and XGBoost in terms of accuracy. Deep learning models, especially LSTM, are more effective than traditional machine learning algorithms for time series forecasting. [9]
- This study uses a data-driven approach with Support Vector Regression (SVR) to predict groundwater levels and water consumption for domestic use. The model incorporates geographical location and climate data to estimate available water resources for various purposes. [10]

Objectives

- **Analyze Historical Groundwater Availability:** Evaluate groundwater availability using historical data .
- **To predict ground water availability:** Applying machine learning to predict the ground water availability
- **Implement Real-Time Water Quality Monitoring:** Deploy IoT sensors to continuously monitor water quality parameters such as temperature, turbidity, TDS, and pH.
- **Visualize Results:** Present findings through graphical representations and reports to support decision-making and resource management.

Problem Statement

To Monitor Real-Time Reservoir Water Pollution Data and to Predict Underground Water Availability for Sustainable Resource Management.

Sensor's Information

pH Sensor

A pH sensor helps to measure the acidity or alkalinity of the water with a value between 0-14

pH is a logarithmic scale that measures how acidic or basic a body of water is.

It's a crucial parameter for assessing water quality because it can affect the chemical and biological processes that occur in water bodies.

Source: <https://shorturl.at/uALS3>

TDS Sensor

The Gravity Analog TDS Sensor has a TDS measurement range of 0–1000 parts per million (ppm) and an accuracy of $\pm 10\%$ FS at 25°C

Total dissolved solids (TDS) in water can be important for water pollution

Because high levels can indicate harmful contaminants, like iron, manganese, sulphate, bromide, and arsenic. TDS can also impact the water's flavour, Odor, and overall palatability.

Source: <https://shorturl.at/epsJ0>

Turbidity Sensor

The range of sensing capacity is 5 - 4000 NTU

Turbidity sensors are used to reduce waste, improve yields, and analyse water quality in a wide range of industries.

For samples with high amounts of TSS and TDS, the difference in the light intensity from the transmission beam is measured to obtain the turbidity result.

While light scattering is more suitable for samples with low amounts of TSS and TDS.

Turbidity sensors use light to detect a solution's turbidity level, it is important to reduce the amount of external light when using the sensor.

Source: <https://tinyurl.com/2ya3ryjk>

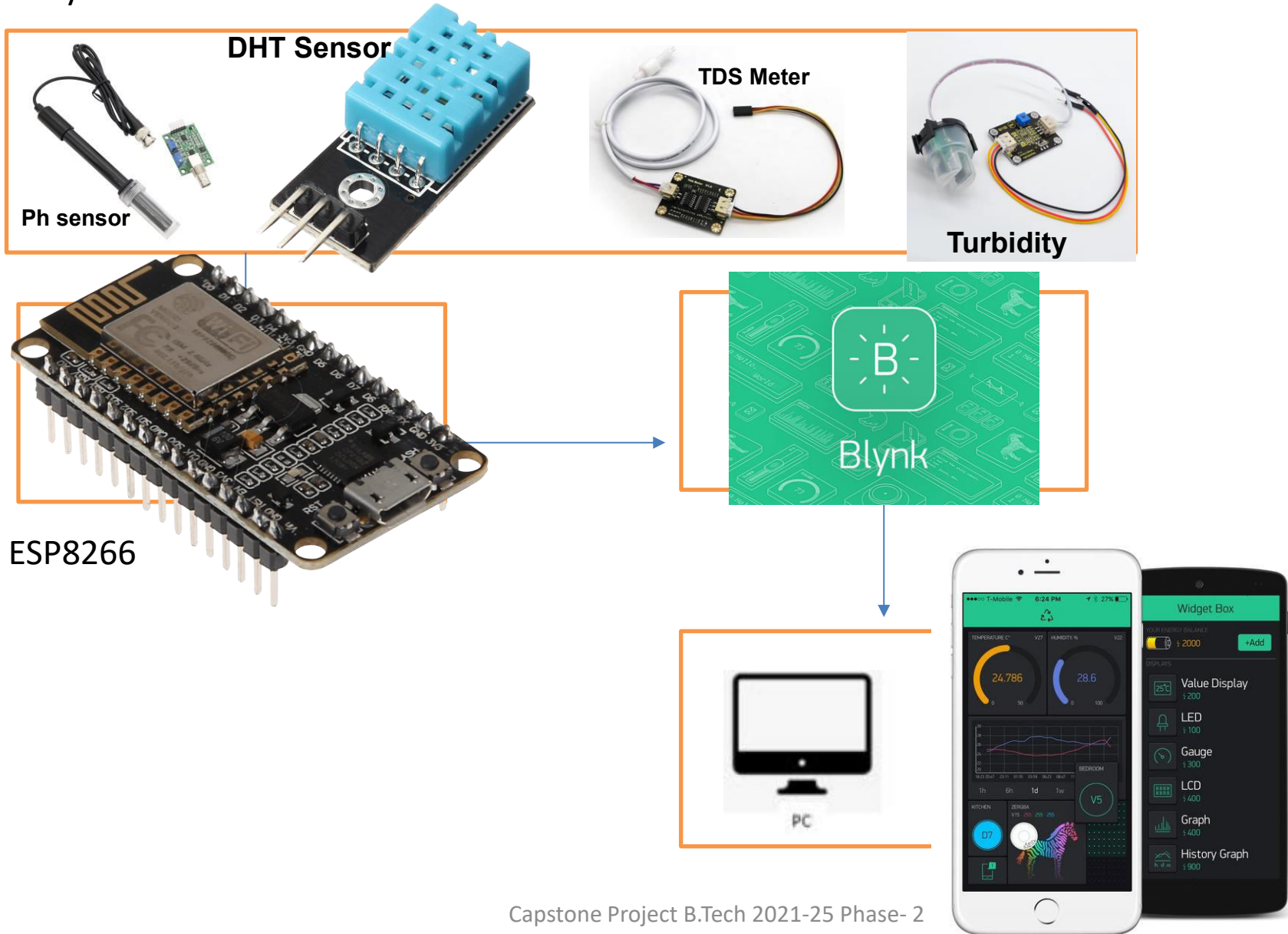
DHT Sensor

The DHT sensor is commonly used to measure both temperature and humidity, with a sensing range for humidity from 20% to 80% RH and temperature from 0°C to 50°C, depending on the model.

DHT sensors are widely utilized in environmental monitoring, HVAC systems, and agricultural applications to ensure optimal conditions by monitoring climate parameters. For accurate humidity measurements, the sensor uses a capacitive humidity sensing component, while a thermistor is used for temperature detection.

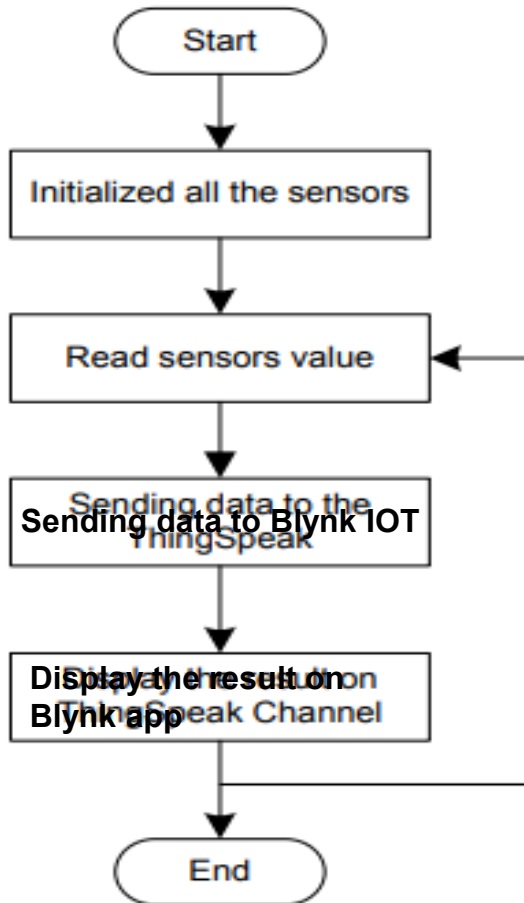
Methodology

System Architecture



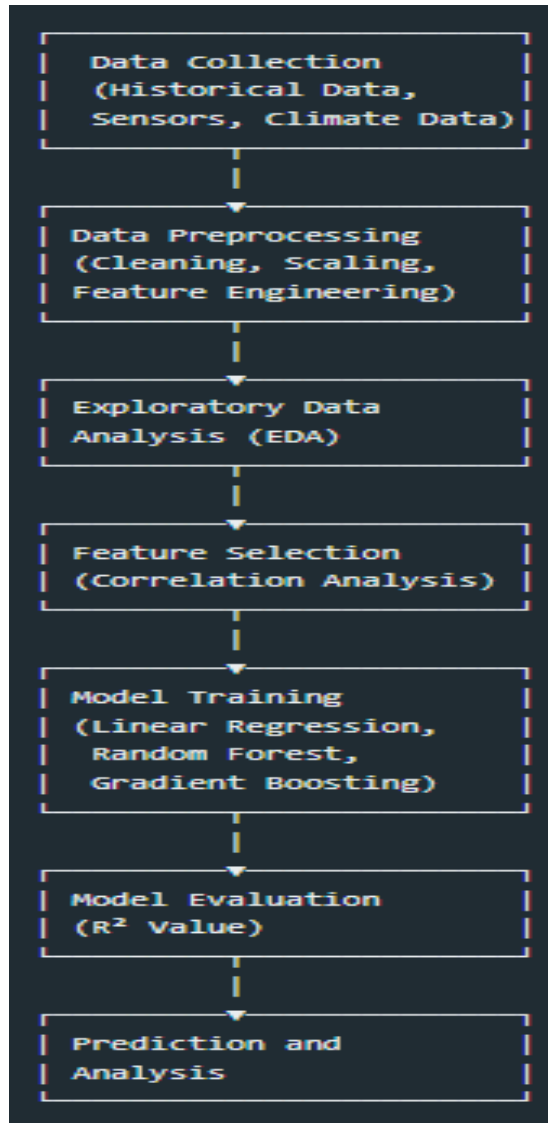
Methodology

Flow Chart of Real time Monitoring Water Pollution



Methodology

Analysis of Ground Water Level



Hardware / Software Implementation

Hardware Components- Turbidity sensor, ESP 32, DHT Sensor , pH sensor, TDS sensor .

Software Components- Blynk IOT , Groundwater dataset , Python.

Dataset: *<https://www.kaggle.com/code/tuhinssam/india-ground-water-exploratory-data-analysis/input>*

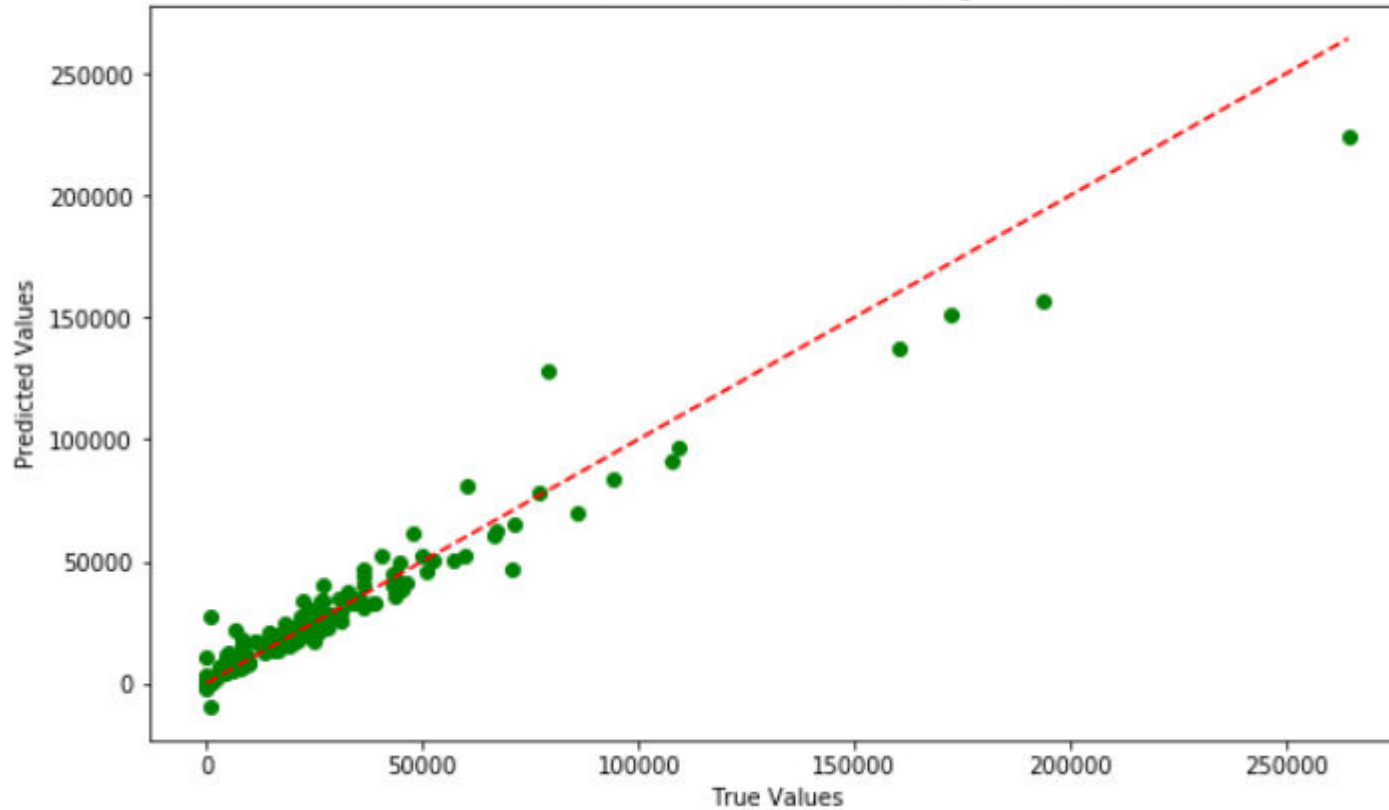
The dataset is provided by <https://data.gov.in/>. The dataset is open for analysis and research.

Results and Analysis

Linear Regression model

Linear Regression R2 Score: 0.9380242645681784

True vs Predicted Values (Linear Regression)

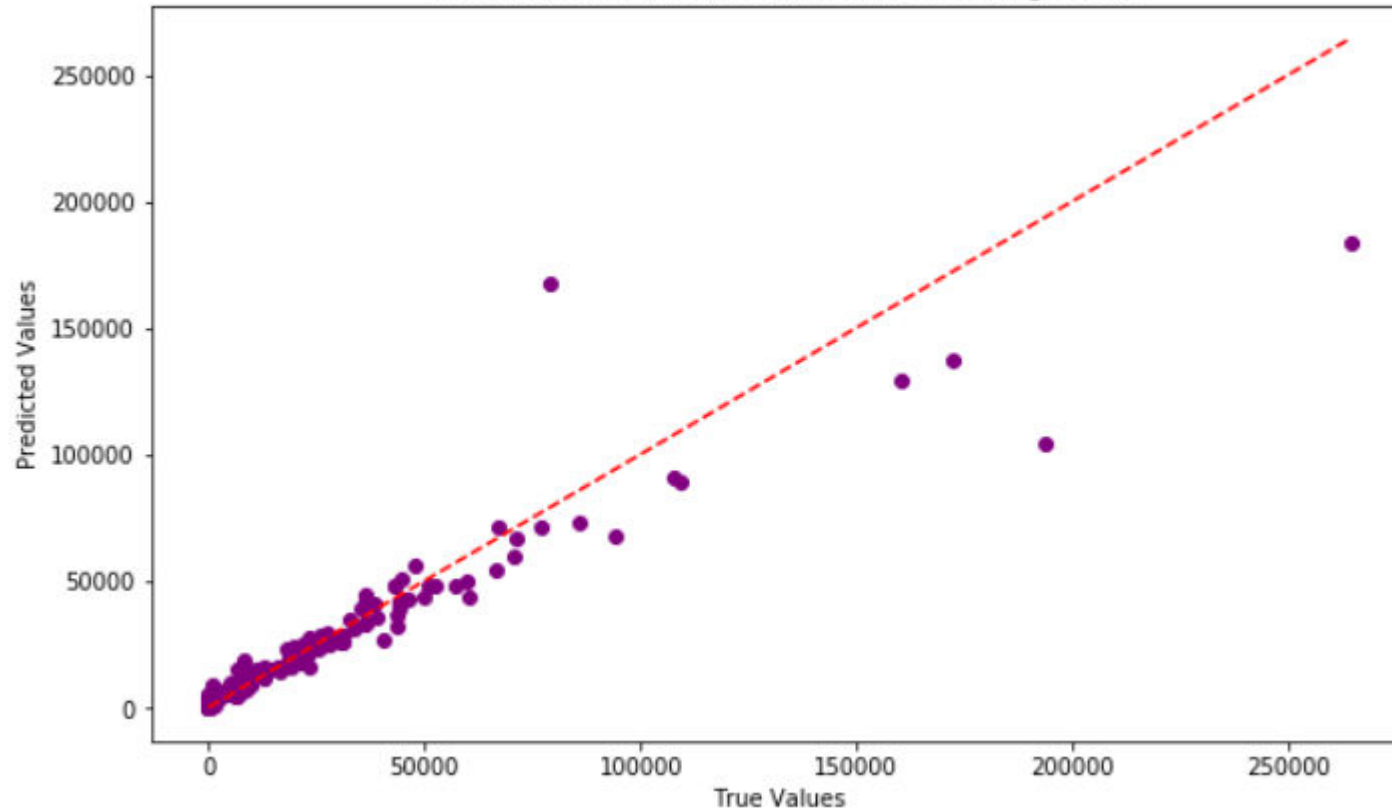


Results and Analysis

Random Forest Regressor

Random Forest Regressor R2 Score: 0.8502557981172599

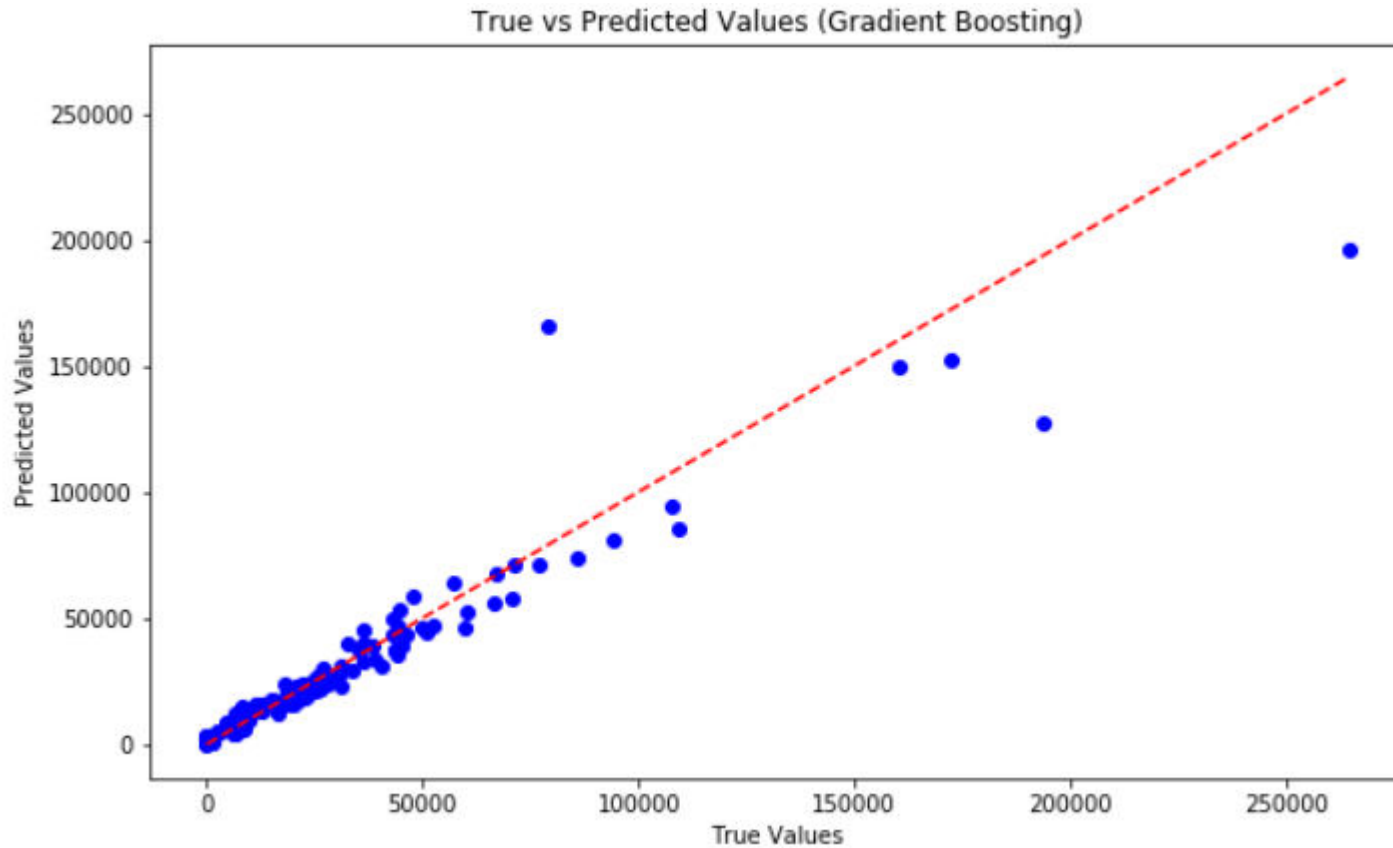
True vs Predicted Values (Random Forest Regressor)

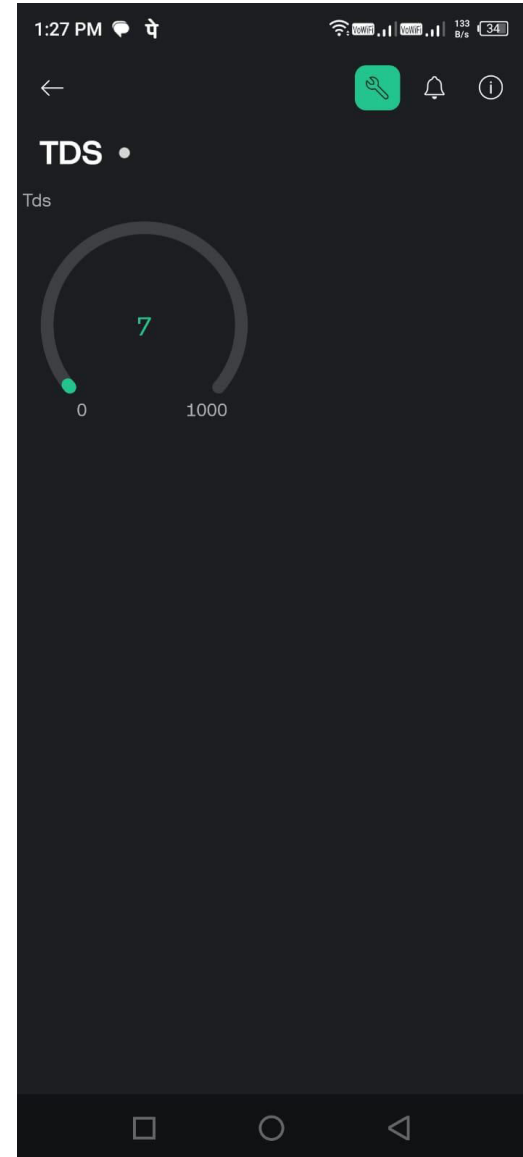
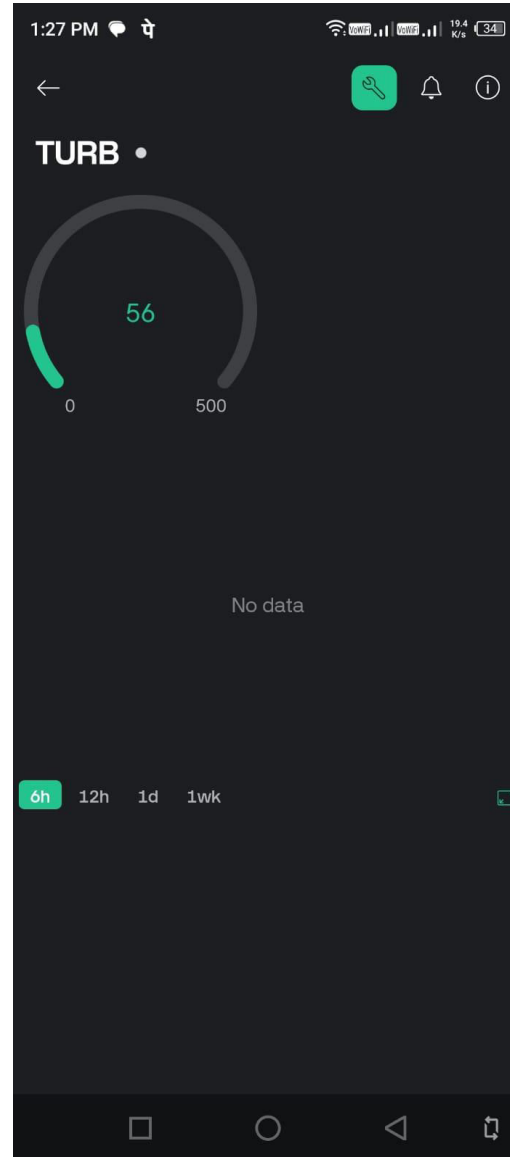
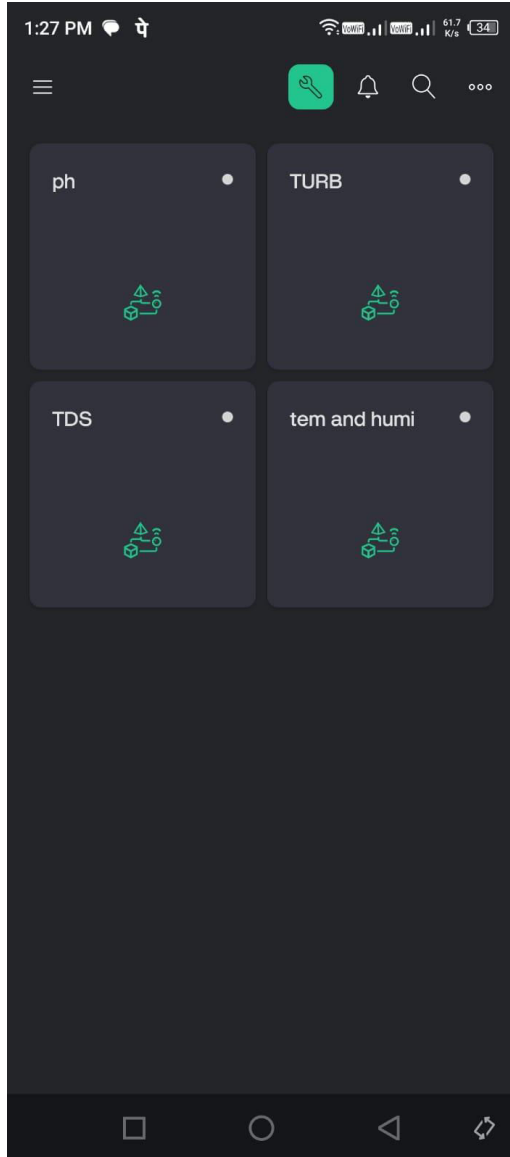


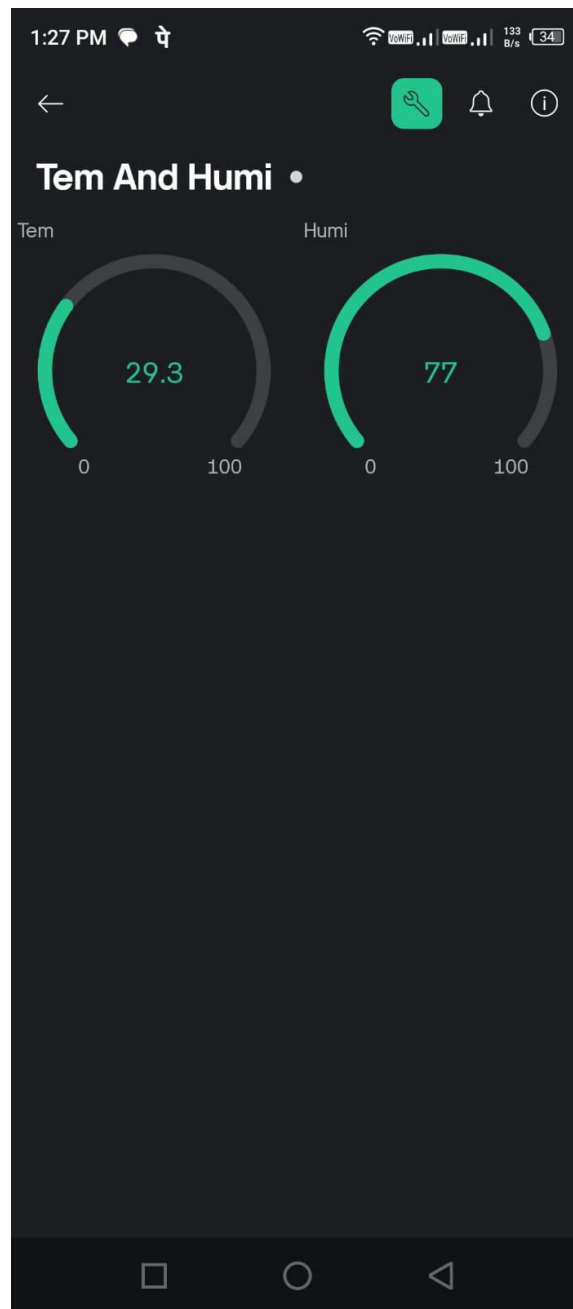
Results and Analysis

Gradient Boosting Regressor

Gradient Boosting Regressor R2 Score: 0.8933798539388684







Results and Analysis

```
#define BLYNK_TEMPLATE_ID "TMPL3mWfugSm-"
#define BLYNK_TEMPLATE_NAME "TURB"
#define BLYNK_AUTH_TOKEN "mCOLfunVMhaFI49ZpyHtFKnedf80Kx-S"
#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "abhikatate"; // Your Wi-Fi name
char pass[] = "bhatru123"; // Your Wi-Fi password

BlynkTimer timer;

// Analog pin for turbidity sensor
#define TURBIDITY_PIN A0 // Connect the turbidity sensor output to A0

// Function to read turbidity and scale the value
int readTurbidity() {
    int rawValue = analogRead(TURBIDITY_PIN); // Read the sensor value
    int scaledValue = map(rawValue, 0, 1023, 100, 0); // Scale to 0-100
    return constrain(scaledValue, 0, 100); // Ensure it's within 0-100
}

// Function to send sensor data to Blynk
void sendSensorData() {
    int turbidityValue = readTurbidity(); // Get the turbidity reading

    // Print turbidity value to Serial Monitor
    Serial.print("Turbidity Value: ");
    Serial.println(turbidityValue);

    // Send scaled turbidity value to Blynk
```



Model	R ² Score
Linear Regression	0.9380
Gradient Boosting Regressor	0.8934
Random Forest Regressor	0.8503

Based on the R² values, **Linear Regression** has the highest score, indicating the best fit for predicting groundwater availability, followed by **Gradient Boosting Regressor** and **Random Forest Regressor**.

Deliverables

- **Historical Groundwater Availability Analysis:** Insights into groundwater availability patterns based on historical data, using machine learning techniques.
- **Real-Time Water Quality Monitoring:** Implementation and evaluation of an IoT-based system for continuous monitoring of water quality parameters such as temperature, turbidity, pH, and TDS.
- **Predictive Insights:** Comprehensive analysis and predictive insights on groundwater resources based on historical trends.
- **Research Paper:** A detailed research paper documenting the methodology, results, and implications of integrating IoT for real-time monitoring with machine learning for groundwater analysis.

References

- [1] K. Banerjee et al., "Assessing Water Quality Index Near Industrial Regions and Aiding in Effective Water Management and Controlling Water Pollution Level," 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), Uttar Pradesh, India, 2022, pp. 1987-1991
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References

- [5] European Journal of Sustainable Development A methodology based on spatial distribution of parameters for understanding affect of rainfall and vegetation density on groundwater recharge Vijai Singhal and Rohit Goyal
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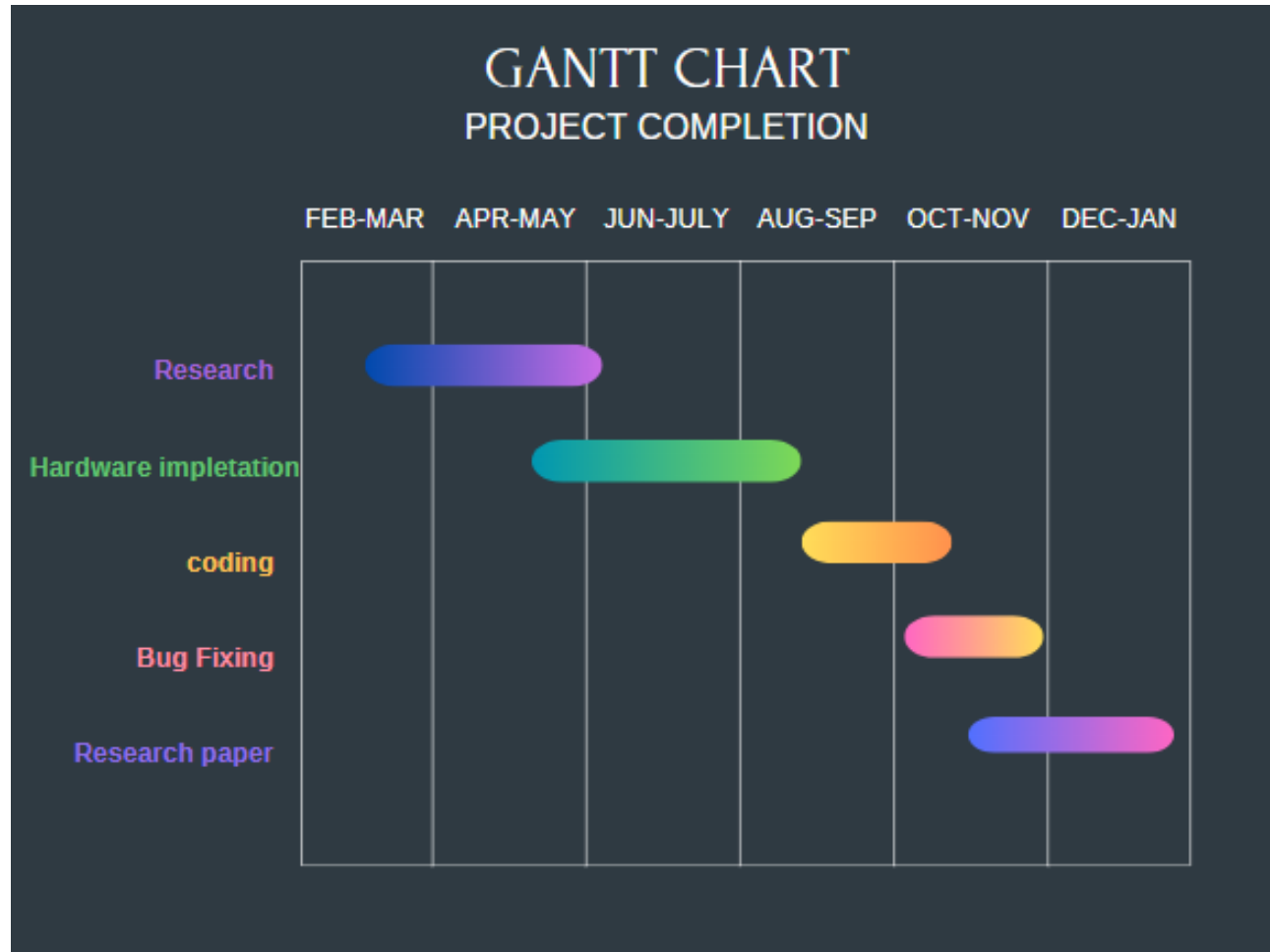
References

- [8] N. Iqbal et al., "Groundwater Level Prediction Model Using Correlation and Difference Mechanisms Based on Boreholes Data for Sustainable Hydraulic Resource Management," in IEEE Access, vol. 9, pp. 96092-96113, 2021, Doi: 10.1109/ACCESS.2021.3094735.
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Summary

- **Real-Time Monitoring:** Live sensor data enables real-time monitoring of water quality, allowing immediate detection and response to pollution events.
- **Precision and Efficiency:** Utilizing live data ensures precise measurements, leading to accurate analysis of water quality and efficient water resource management.
- **Pollution Control:** Live sensor data helps identify pollution sources and levels, facilitating targeted interventions to mitigate environmental impacts.
- **Groundwater Availability:** Understanding groundwater resource dynamics, assessing impacts of climate change, and supporting sustainable water management.

Project timeline



Individual Contributions

Akash C K
PES1UG21EC022

Presentation No.	Individual Contribution
1.	Sensor Documentation and Codes: Gathered detailed documentation for various sensors used in the water quality monitoring system, including specifications and functionalities. Research on ML Models: Collected information about machine learning models applicable for predicting groundwater availability. Integration of Topics: Assisted in aligning the water quality monitoring and groundwater prediction aspects into a cohesive project framework.
2.	Resolving IoT Data Transfer Issues: Troubleshoot and worked on resolving challenges related to data transfer between IoT sensors and the central system. Finalizing ML Models: Finalized the machine learning models for groundwater availability prediction, selecting the most suitable algorithms. Documentation and Code Gathering: Consolidated necessary codes and relevant documentation for implementing machine learning workflows.
3.	Water Quality Monitoring Implementation: Successfully implemented the code for real-time water quality monitoring using sensors. Groundwater Prediction Models: Executed the machine learning models (Linear Regression, Random Forest, Gradient Boosting) for predicting groundwater availability. System Integration and Testing: Ensured seamless integration of both modules (sensor-based monitoring and ML-based prediction) and validated their performance.

Individual Contributions

Akash Ravi Bhat
PES1UG21EC025

Presentation No.	Individual Contribution
1.	I and my teammates finalized our project problem statement, and we listed hardware components that are required for our project. We went to the electronics components store, and we bought some materials and components required for our project. I started to research the dataset for our machine learning project. We got the dataset from the government website and performed exploratory data analysis. We applied the linear regression model for our groundwater analysis model.
2.	I and my teammates combined all four sensors, and we analysed them with different IoT cloud platforms like ThingSpeak and Blynk IoT, and then I found Blynk IoT is suitable for our project. So, we combined all four pH, turbidity, TDS, and temperature sensors with the ESP8266 and collected data, and we applied the random forest and gradient boosting methods along with linear regression methods, which we applied initially and demonstrated hardware and software models in our ISA2 presentation.
3.	I met our guide after ISA and took the project diary and filled it out accordingly. I did the project poster in Canvas and did that with necessary photos and required fillings. Also, I and Tejas started to build a report, and Abhishek and Akash Katate started to do the PPT. We finalized our report, and I gave it to plagiarism checks. I got a good result in both plagiarism and AI content checks, and we settled up all the hardware components along with the necessary platforms, and we finalized our machine learning model with 3 models to demonstrate, and we divided the presentation topics for the final presentation and were well prepared for our ESA.

Individual Contributions

Abhishek Shetty
PES1UG21EC008

Presentation No.	Individual Contribution
1	* I collected details of IoT sensors for water quality monitoring, focusing on specifications. * Analyzed groundwater availability datasets to identify key features and trends. * Combined sensor details with dataset insights to structure and streamline project components effectively.
2	* I collected and documented IoT sensors for water quality monitoring, detailing their specifications and deployment requirements to ensure suitability for the project. * Conducted an in-depth analysis of datasets to identify key features, trends, and patterns related to groundwater availability, aiding in data-driven decision-making. * Additionally, contributed to structuring the project by integrating comprehensive sensor documentation with initial dataset findings.
3	* Successfully implemented the hardware setup for the IoT-based water quality monitoring system, ensuring smooth and reliable operation. * Assisted in applying machine learning models to predict groundwater availability, data-driven insights for enhanced accuracy. * Contributed extensively to the final project documentation, covering technical and functional details to provide a clear and detailed account of the project's scope and achievements. * Well prepared for ESA presentation and model demonstration

Individual Contributions

Tejas V P

PES1UG21EC910

Presentation No.	Individual Contribution
1.	I contributed in finalizing the project and helping to narrow down the solution and we started doing research on that topic, there I learned a lot regarding the causes of water pollution and what are the problems present in The current system , so form there we took that problem statement and red some research papers on that topic , from there we thought of doing hardware project and ml project , for that I gone through research papers and journals
2.	After we finalized our requirements we learned about sensors necessary for this project . we bought it from the store and started setting up the hardware sensors , we gone through some errors while uploading the code to sensors and we got some wrong output values , then we gone through the necessary websites which have solutions for this issues , then we implemented it to our project
3.	I contributed to developing machine learning code for predicting groundwater availability. Additionally, I was actively involved in creating presentations, drafting detailed reports, debugging issues, and finalizing the hardware aspects of the project.

Deliverables Submissions

Project ID : W11		
Sl. No.	Submission	Details
1	Project Report	As per format with Plagiarism report and Poster
2	Project Poster	A3 size color hard copy and soft copy
3	Project Diary	Submitted to the Panel
4	Individual Contribution Form	Submitted and signed by each student

Real-Time IoT-Enabled Water Quality Monitoring and Groundwater Availability Analysis Using Machine Learning

Abhishek A Shetty , Akash Chanabasu Katate , Akash Ravi Bhat , Tejas V P

Project guide : Prof. NAGARAJ L J

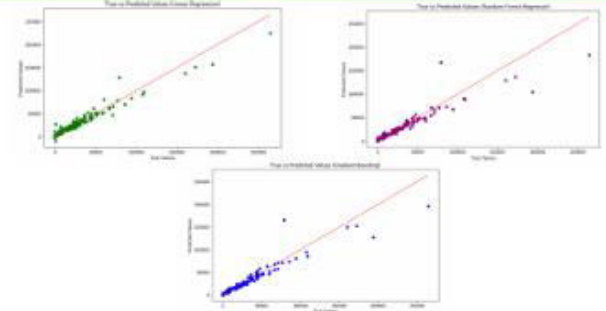
Motivation

This motivation highlights the urgency of addressing water pollution, which poses significant threats to environmental health, public well-being, and sustainable development. With rising pollution levels driven by industrialization, urbanization, and agriculture, traditional monitoring methods are inadequate for providing real-time data. This project addresses the critical need for continuous monitoring, enabling rapid responses to pollution and informed decision-making to protect water resources.

Theory

Groundwater availability is crucial for sustainable resource management, requiring a combination of historical data analysis, real-time monitoring. Machine learning models can analyze trends and forecast groundwater levels, while IoT sensors continuously track water quality parameters like pH, turbidity, and TDS. By integrating these technologies and visualizing results through dashboards and reports, decision-makers can effectively manage groundwater resources, mitigate over-extraction risks, and ensure water quality for various uses.

Analysis



Implementation

Hardware Implementation

The project uses sensors like turbidity, pH, TDS, and DHT to monitor water quality and environmental conditions. These sensors are crucial for measuring clarity, acidity, mineral concentration, temperature, and humidity.

An ESP32 microcontroller integrates these sensors and connects to the cloud via Wi-Fi. This setup enables real-time data transmission and ensures efficient remote monitoring. The system is designed to operate reliably in diverse environmental conditions, ensuring consistent performance.

Software Implementation

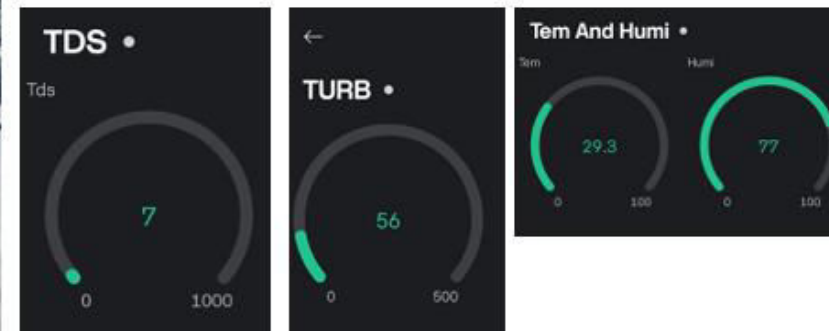
The Blynk IoT platform provides real-time visualization of sensor data and sends alerts for unsafe water parameters. It ensures accessible and actionable data for stakeholders.

Python programming is used for data cleaning, analysis, and machine learning on groundwater datasets. Models like Linear Regression and Random Forest predict groundwater availability for sustainable management.



Results and Conclusion

This project will effectively illustrate integration of real-time water quality monitoring through IoT enabled by machine learning predictive analysis on the basis of groundwater availability. The precise and efficient tracking of essential water quality parameters like turbidity, pH, temperature, and TDS will be assured with advanced sensors and cloud-based platforms.. With these machine learning techniques applied to historical data, predictions regarding the availability of groundwater were accurate enough to greatly aid in sustainable water resource management.



Q & A

Thank You